SIMATS ENGINEERING

ASSESSMENTTOOL1:ShortAnswerTest

COURSE NAME: Cloud Computing and Big Data Analytics ***COURSE CODE:*** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) ***Total Marks: 25***
Question Count: 5

Code of Conduct

I Lochana Sri R.S.(192521348)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Lochana Sri R.S.

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of ***data centers*** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

ShortAnswer Test Rubric

Criteria	***Weig htag e***	***Level 4 - Exemplary***	***Level 3 - Proficient***	***Level 2 - Developing***	***Level 1 - Beginning***
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1: ASSESSMENT - 1.

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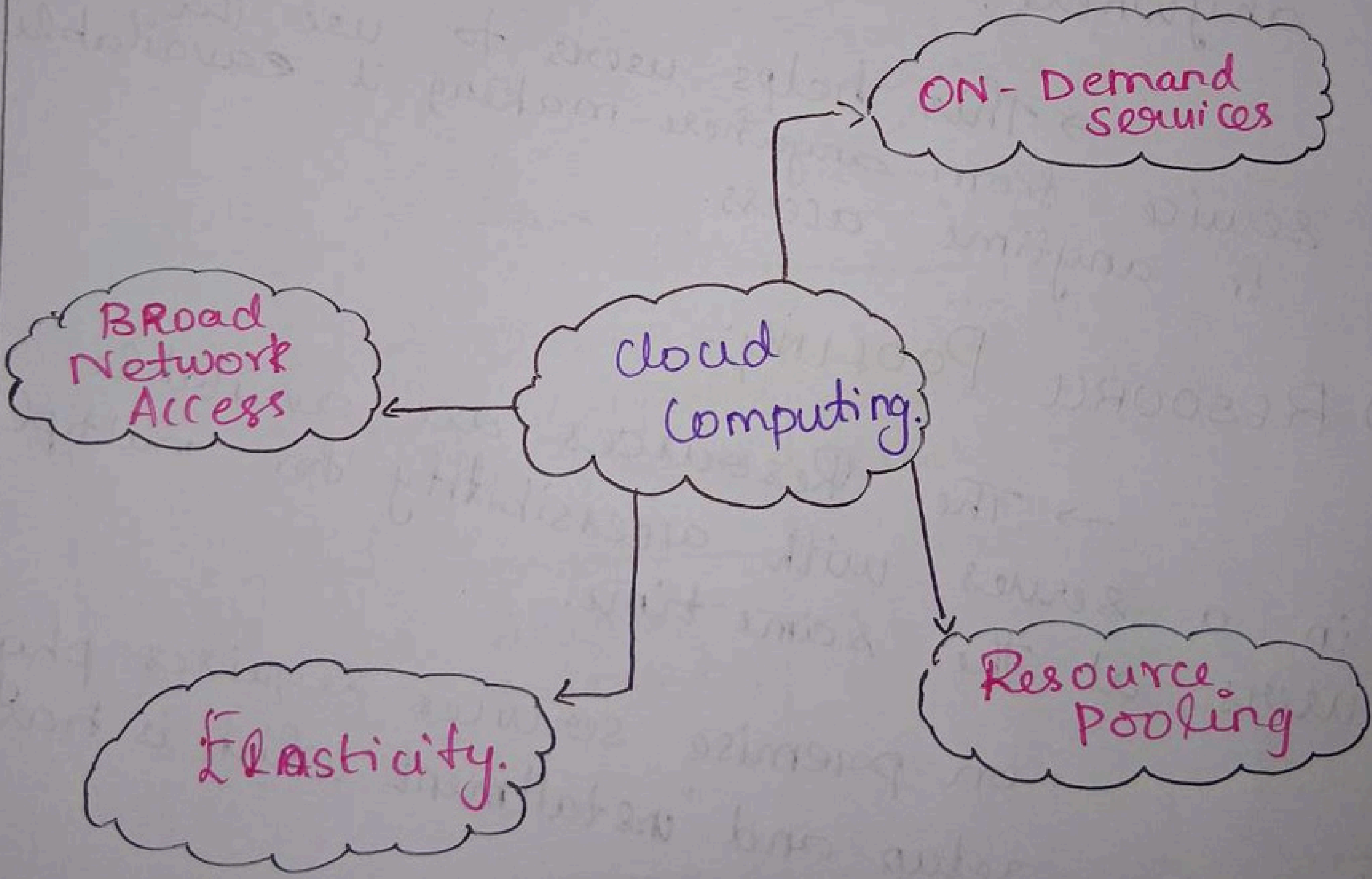
COURSE: Cloud Computing
CODE : CSA15

1. CLOUD COMPUTING:-

cloud computing is a on-demand service which provides servers, databases, storage, network and etc on a pay as-you-go method.

Eg:- Aws, Azure, ~~et~~ Google cloud.

CHARACTERISTICS:-



i) ON DEMAND SERVICES:-

→ Users can interact with the cloud service provider without having a human interaction.

→ This gives effieciency and easy to access.

ii) BROAD NETWORK ACCESS:-

→ The Network is available all over the places, so that users can access and use internet ~~eithe~~ from anywhere.

→ This helps users to use the available service from anywhere making it for anytime access.

iii) RESOURCE POOLING:-

→ The Resources are available in a server with accessibility for multiple users at the same time.

→ on premise services requires physical hardware setup and installment which is hard & costly.

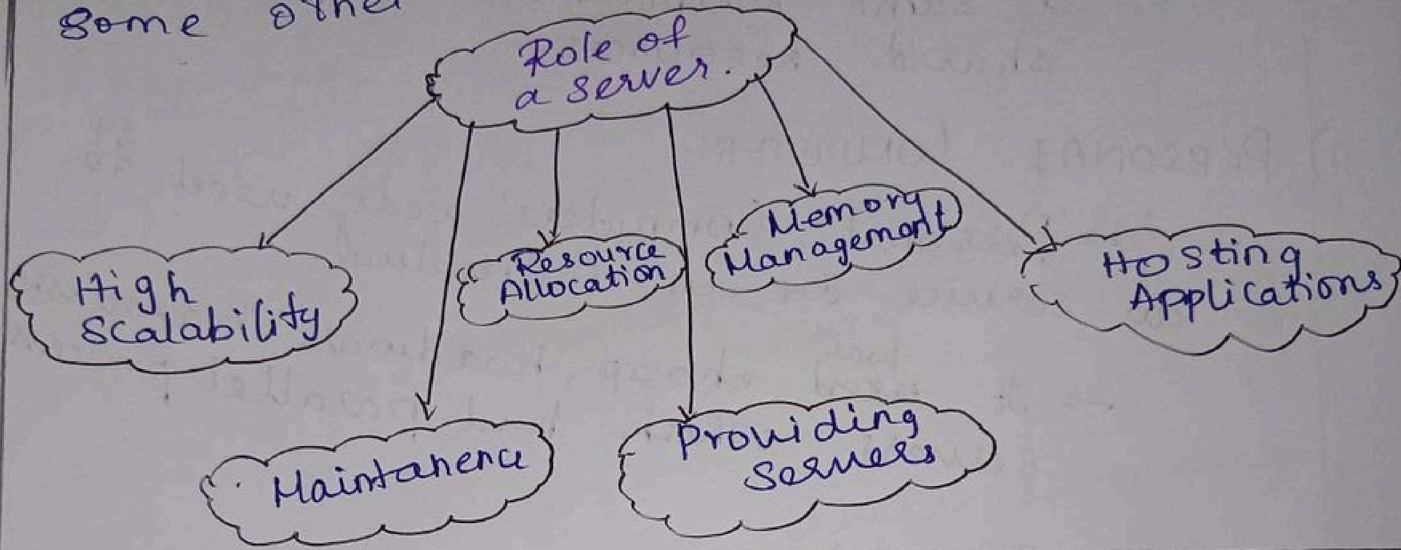
iv) ELASTICITY:-

→ Rescources can be scaled easily with days on demand with more servers.

→ Making this cost effecient and highly scalable.

VIRTUALIZATION:-

Virtualization is the process of creating virtual machines with separate resources, OS, network for its own in some other computer to access services using a hypervisor.



CLOUD COMPUTING	VIRTUALIZATION.
→ On demand service which provides resources, OS, network on a pay-as-you-go basis.	→ Creating a virtual environment for a Guest host to run
→ Uses virtualization to provide software and services	→ Uses a hypervisor to control and manage.
→ It is available and accessible from anywhere using just an internet	→ Needs to be installed and set up in a Computer.
Eg:- AWS, Microsoft Azure	Eg:- VMware.

2. EVOLUTION OF CLOUD COMPUTING:-

i) MAINFRAMES:-

→ Mainframes where the most commonly used computers around 18th century

→ This is where many people used a same Mainframe, allowing for shared resources.

ii) PERSONAL COMPUTER:-

→ Personal Computers were used to do service for an individual.

→ It ~~used~~ was cheap, hardware based x86 servers which had parallel processing.

iii) CLIENT-SERVER:-

→ client - server is the one where a provider and user is ther.

→ The client requests for resources from the main server which provides.

iv) CLOUD COMPUTING:-

→ cloud computing is where virtualization comes in, An guest OS host can be installed virtually in a different OS based computer.

→ This reduces the cost, increases efficiency and provides efficient storage management.

4. DATA CENTERS:

Data Centre is a place where all the servers with all its resources are being stored in.

ROLE OF DATA CENTER:-

i) PROVIDING SOURCES:-

It allocates and provides resources according to user requests.

ii) STORAGE:-

It stores large no. of files in the same place providing easy access.

iii) HIGH SCALABLITY:

It accomades servers according to the demand of servers.

5.

MODEL	DEFINITION	ROLES	EXAMPLE.
IAAS	Infrastructure as a service: provides the OS, Hardware for the users to use.	Focuses mainly on allowing the users to access the infrastructure	~~Google~~ EMC2. Amazon
PAAS	Platforms As A Service: Provides platform as as services for the users to develop applications by providing network and suitable environment.	Mainly focuses on providing resources to develop Application.	Google.
SAAS	Software as a Service: Provides application software to users for using without installation.	Focuses on mainly providing software for the users to use.	Ready to use softwares on internet.